

SECTION 1: DATA QUALITY CHECKS

A1. Invalid Stays

Answer: 120 bookings have invalid stays (checkout date $\leq$ check-in date)

Method: Filtered rows where checkout_date $\leq$ checkin_date after converting both columns to datetime. These represent data entry errors and were removed before all further analysis.

A2. Review Rating Analysis by Segment

RAW (before normalization):

Corporate → min: 3.0 max: 10.0 mean: 7.25 count: 1,287

Group → min: 1.0 max: 5.0 mean: 3.77 count: 590

Individual → min: 1.0 max: 5.0 mean: 3.77 count: 3,269

NORMALIZED (Corporate $\div 2$ to match 1–5 scale):

Corporate → mean: 3.62

Group → mean: 3.77

Individual → mean: 3.77

One-line implication:

Corporate ratings appear highest on raw data (avg 7.25 on 1–10 scale) but after normalizing to the same 1–5 scale, Corporate (3.62) is actually the LOWEST rated segment — the dual scale creates a false impression of superior Corporate satisfaction.

A3. Realized Revenue — Luxury Properties

Answer: ₹90,694,052.93

Footnotes applied:

- Footnote 8: Only Completed bookings counted (not Cancelled/No-Show)
- Footnote 1: Invalid stays excluded
- Footnote 3: Zero-room bookings excluded

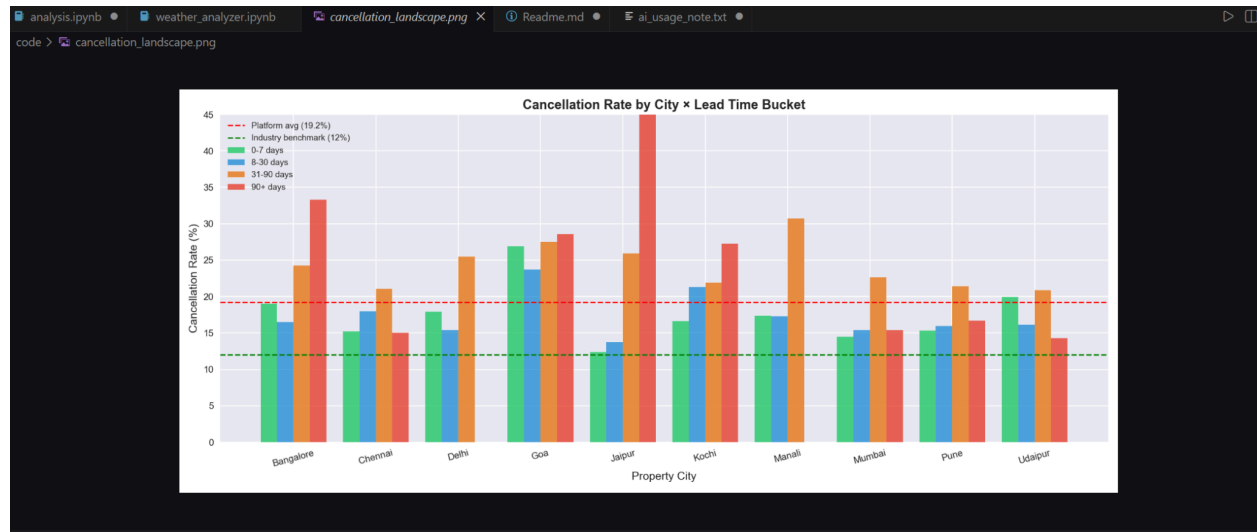
Wrong answer without footnotes: ₹1,17,431,484.47

Overstatement prevented: ₹26,737,431.54

SECTION 2: CANCELLATION CASE STUDY

Platform cancellation rate: 19.2% (benchmark: 12%, target: 14%)

A1. Cancellation Landscape



Key finding: Lead time is the strongest driver of cancellation variance. The 90+ day bucket shows elevated rates across almost all cities. Jaipur and Bangalore show the highest red bars (90+ days) but with very low booking volumes. The 31–90 day bucket (orange) is consistently above the platform average red dashed line across all cities.

A2. Rate vs Volume

TOP 3 BY CANCELLATION RATE:

1. Jaipur | 90+ days → 100.0% (1 booking — statistically meaningless)
2. Bangalore | 90+ days → 33.3% (15 bookings)
3. Manali | 31-90 days → 30.7% (218 bookings) ← TRUE worst slice

TOP 3 BY CANCELLATION VOLUME:

1. Chennai | 8-30 days → 18.0% rate, 125 cancellations
2. Pune | 31-90 days → 21.4% rate, 121 cancellations
3. Chennai | 31-90 days → 21.1% rate, 120 cancellations

Why this distinction matters:

The board's 5pp target requires eliminating ~609 cancellations platform-wide ($11,821 \times 5\% = 591$). High-rate but low-volume slices

like Jaipur (1 booking) contribute negligibly. The highest-volume slices (Chennai, Pune) have moderate rates but far greater impact potential. Targeting rate alone is a trap — volume is what moves the platform number.

A3. Root Cause: Hypothesis Testing

Target slice: Manali | 31–90 days (30.7% cancel rate, 218 bookings)

HYPOTHESIS 1 — Lead Time Effect:

31-90 day cancel rates across all cities:

Manali: 30.7% Goa: 27.5% Jaipur: 25.9% Delhi: 25.5%

Bangalore: 24.2% Mumbai: 22.7% Udaipur: 20.9%

Verdict: PARTIAL — lead time elevates rates everywhere but Manali is still 3.2pp above the second worst city. Not fully explained.

HYPOTHESIS 2 — Channel Mix Effect:

OTA share: Manali 38.2% vs Platform 35.2% (only +3pp difference)

OTA cancel rate platform-wide: 22.0%

Verdict: RULED OUT — a 3pp higher OTA share cannot explain an 11.8pp gap above platform average. Mathematically too small.

HYPOTHESIS 3 — City + Season Effect:

Manali October cancel rate: 31.7%

Mumbai October cancel rate: 16.7%

Manali July cancel rate: 13.8% (lowest month)

Verdict: SUPPORTED — October spike is unique to Manali, matching Rohtang Pass road closures due to early snowfall. People book Manali months ahead for October travel then cancel when mountain roads close.

A4. Targeted Recommendation

WHO: Manali bookings with 31–90 day lead time

(218 bookings, 30.7% cancel rate — highest meaningful slice)

WHAT: Opt-in 20% non-refundable deposit at booking time + 8% total discount as incentive. Remaining 80% charged 7 days before check-in. Targets root cause — financial commitment reduces speculative bookings for uncertain October mountain travel.

QUANTIFIED IMPACT:

Current cancellations in slice : 67
Expected savings (30% reduction): 20 cancellations
Expected savings (50% reduction): 34 cancellations
New platform rate (conservative) : 19.0%
New platform rate (optimistic) : 18.9%
Expected reduction : 0.2pp to 0.3pp
Board target coverage : 4% to 6% of 5pp target

HONEST CONTEXT: Manali 31-90 days = 1.8% of platform bookings.
Full 5pp target requires similar policies across Chennai (125 cancellations) and Pune (121 cancellations) — combined potential additional impact of 1.5–2.0pp.

RISK: Customers may choose OTA competitors offering free cancellation. Mitigation — keep deposit as opt-in with visible discount, never as a mandatory gate.

SECTION 3: SQL SCHEMA + QUERIES

SCHEMA DESIGN

```
CREATE DATABASE hotel_analysis;
USE hotel_analysis;
USE hotel_analysis;

CREATE TABLE customers (
    customer_id      INTEGER      PRIMARY KEY,
    customer_name    VARCHAR(100) NOT NULL,
    customer_segment VARCHAR(20)  NOT NULL,
    signup_date      DATE         NOT NULL,
    home_city        VARCHAR(50),
    loyalty_tier     VARCHAR(20)  NOT NULL
);

CREATE TABLE properties (
    property_id      INTEGER      PRIMARY KEY,
    property_name    VARCHAR(100) NOT NULL,
    property_city    VARCHAR(50)  NOT NULL,
    property_type    VARCHAR(20)  NOT NULL,
    star_rating      INTEGER      NOT NULL,
    total_rooms      INTEGER      NOT NULL,

    UNIQUE (property_name, property_city)
);

CREATE TABLE bookings (
    booking_id       INTEGER      PRIMARY KEY,
    customer_id      INTEGER      NOT NULL,
    property_id      INTEGER      NOT NULL,
    booking_date     DATE         NOT NULL,
    checkin_date     DATE         NOT NULL,
    checkout_date    DATE         NOT NULL,
    room_type        VARCHAR(30),
    num_rooms        INTEGER      NOT NULL,
    nights           INTEGER      NOT NULL,
    booking_channel  VARCHAR(30)  NOT NULL,
    adr              DECIMAL(10,2) NOT NULL,
    discount_amount  DECIMAL(10,2) NOT NULL DEFAULT 0,
    coupon_code      VARCHAR(20),
    total_amount     DECIMAL(10,2) NOT NULL,
```

```

payment_method    VARCHAR(30),
booking_status    VARCHAR(20)    NOT NULL,

FOREIGN KEY (customer_id) REFERENCES customers(customer_id),
FOREIGN KEY (property_id) REFERENCES properties(property_id),

CHECK (checkout_date > checkin_date),
CHECK (num_rooms > 0)
);

CREATE TABLE reviews (
    review_id       INTEGER        PRIMARY KEY AUTO_INCREMENT,
    booking_id      INTEGER        NOT NULL UNIQUE,
    review_rating    DECIMAL(3,1)  NOT NULL,
    review_date      DATE           NOT NULL,

    FOREIGN KEY (booking_id) REFERENCES bookings(booking_id),
    CHECK (review_rating >= 1 AND review_rating <= 10)
);

```

QUERY 1: Top Revenue Property Per City — RANK()

```

USE hotel_analysis;
-- Top revenue property per city using RANK()
WITH property_revenue AS (
    -- revenue per property
    SELECT
        p.property_id,
        p.property_name,
        p.property_city,
        SUM(b.total_amount) AS total_revenue
    FROM bookings b
    JOIN properties p
        ON b.property_id = p.property_id
    WHERE b.booking_status = 'Completed'    -- Footnote 8
    GROUP BY p.property_id, p.property_name, p.property_city
),

ranked AS (
    -- Rank properties within each city by revenue

```

```

SELECT
    property_id,
    property_name,
    property_city,
    total_revenue,
    RANK() OVER (
        PARTITION BY property_city
        ORDER BY total_revenue DESC
    ) AS revenue_rank
FROM property_revenue
)
-- top ranked property per city
SELECT
    property_city,
    property_id,
    property_name,
    total_revenue
FROM ranked
WHERE revenue_rank = 1
ORDER BY property_city;

```

QUERY 2: Frequent Customers — LAG()

```

WITH ordered_bookings AS (
    -- bookings ordered by customer + date
    SELECT
        customer_id,
        checkin_date,
        LAG(checkin_date) OVER (
            PARTITION BY customer_id
            ORDER BY checkin_date
        ) AS previous_checkin
    FROM bookings
    WHERE booking_status = 'Completed'
),
gaps AS (
    -- Calculate gap in days between consecutive bookings

```

```

SELECT
    customer_id,
    DATEDIFF(checkin_date, previous_checkin) AS gap_days
FROM ordered_bookings
WHERE previous_checkin IS NOT NULL -- skip first booking (no
previous)
),

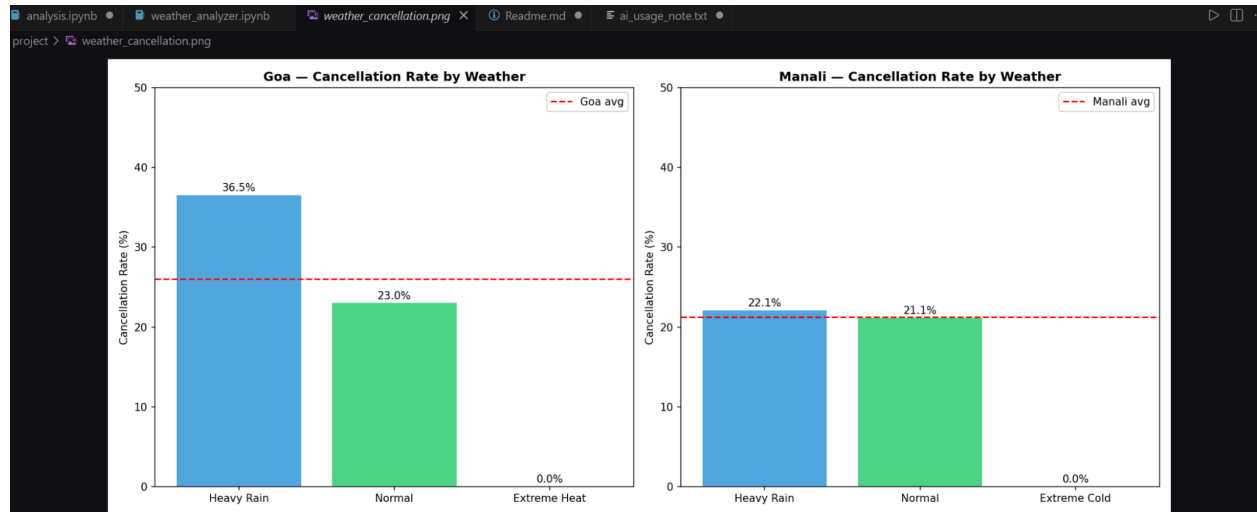
customer_avg_gaps AS (
    -- Average gap per customer
    SELECT
        customer_id,
        AVG(gap_days) AS avg_gap_days,
        COUNT(*) AS num_gaps
    FROM gaps
    GROUP BY customer_id
)
-- customers with average gap under 30 days
SELECT
    COUNT(*) AS frequent_customers,
    ROUND(AVG(avg_gap_days), 1) AS avg_gap_among_frequent
FROM customer_avg_gaps
WHERE avg_gap_days < 30;

```


SECTION 4: WEATHER-CANCELLATION ANALYZER

API: Open-Meteo Historical Weather (ERA5) — free, no key required

Endpoint: <https://archive-api.open-meteo.com/v1/era5>



KEY INSIGHT

Heavy rainfall on check-in day is strongly associated with higher cancellation rates in Goa (36.5%) compared to normal weather days (23.0%) — a 13.5 percentage point difference across 211 rainy-day bookings. This effect is almost entirely absent in Manali, where heavy rain days (22.1%) and normal days (21.1%) show nearly identical rates. This reveals a destination-type dependency: weather-driven cancellations concentrate in beach destinations where weather defines the experience, not mountain destinations where adverse weather is anticipated. A practical implication — weather-indexed flexible cancellation policies (free cancellation if precipitation > 10mm on check-in day) could reduce Goa's cancellation rate while improving customer trust.